

# Joint Intent Classification and Slot Filling

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## 1. Introduction

This report covers the two intent classification and slot filling exercises on the ATIS dataset, evaluated by intent accuracy and slot F1 with the conll scorer. In Part 2.A a decoder-only GPT-2 was built from scratch and given two output heads, then tuned one hyperparameter at a time from a small baseline. It reaches slot F1 87.40 and intent accuracy 89.81. In Part 2.B two pre-trained backbones were fine-tuned in the same multi-task setting: GPT-2, a causal decoder, and BERT-base, a bidirectional encoder. BERT reaches slot F1 95.75 and intent accuracy 97.54, GPT-2 reaches 92.73 and 96.35. Comparing the three models separates the contribution of pretraining from that of bidirectionality.

## 2. Implementation details

**Part 2.A.** The from-scratch model reuses the GPT-2 decoder from the language modeling project with two linear heads on top: a slot head applied at every token position, and an intent head. Because the model is causal, the intent token cannot sit at the start of the sentence, where it would attend to nothing; it is appended at the end, so it sees the whole sequence before predicting the intent. The two cross-entropy losses are summed. Tuning follows the same incremental protocol as the language model: from a small baseline ( $d_{\text{model}}=20$ , one head, one layer, feed-forward width 20), the learning rate is set first, then each capacity hyperparameter is changed one at a time and kept only when development performance improves. Model width and feed-forward width are the changes that help; an extra layer and a dropout layer are reverted because they do not. The final model is wide rather than deep ( $d_{\text{model}}=200$ , four heads, one layer, feed-forward width 512). The main difficulty was a collapse to the majority class, described in the results.

**Part 2.B.** Both backbones are fine-tuned with the same two heads. The challenge is sub-tokenization: the tokenizers split words into several pieces, but slot labels are defined per word. Following the alignment scheme in [1], each word's slot label is placed on its first sub-token, while continuation pieces and special tokens are set to the ignore index so they do not contribute to the loss; at evaluation one slot per word is read back from the first sub-token and passed to the conll scorer. The two architectures need different intent handling: BERT reads the intent from its [CLS] token, while GPT-2, which has none, reads it from the last real token. GPT-2 also requires the fast tokenizer with prefix spaces enabled and its padding token set to the end-of-sequence token.

## 3. Results

All models are evaluated by intent accuracy and slot F1 with the conll scorer on the ATIS test set. Each result is the mean over five seeds with its standard deviation, and the saved checkpoint is the best-development seed. Table 1 and Figure 1 report

the three models: the from-scratch model reaches slot F1 87.40 and intent accuracy 89.81, fine-tuned GPT-2 reaches 92.73 and 96.35, and BERT reaches 95.75 and 97.54.

The main obstacle in Part 2.A was a repeated collapse to the trivial solution, where the model labelled every token  $\emptyset$  and predicted the single most frequent intent. The signature was unmistakable: slot F1 fell to zero while intent accuracy froze at exactly the base rate of the majority class on the development set. The fix was counterintuitive, a higher learning rate rather than a lower one; low rates could not escape the basin, and learning-rate warmup made it worse.

Figure 2 separates the two sources of gain over the from-scratch model. Holding the architecture causal and adding only pretraining, the move to fine-tuned GPT-2, gives +5.3 slot F1 and +6.5 intent accuracy. Holding pretraining fixed and adding bidirectionality, the move from GPT-2 to BERT, gives +3.0 slot F1 but only +1.2 intent accuracy. The asymmetry is informative: bidirectionality helps slot filling far more than intent classification, because slot filling labels each token and benefits from the right-hand context that only a bidirectional encoder sees, while intent is a single whole-sentence decision that the causal model's final token already captures.

As in the language modeling project, the reported numbers carry CUDA nondeterminism of a few tenths, so each saved value is the one whose weights were checkpointed.

*Use of AI tools.* Claude was used minimally, for debugging and refinement.

## 4. References

- [1] Q. Chen, Z. Zhuo, and W. Wang, "BERT for joint intent classification and slot filling," *arXiv preprint arXiv:1902.10909*, 2019.

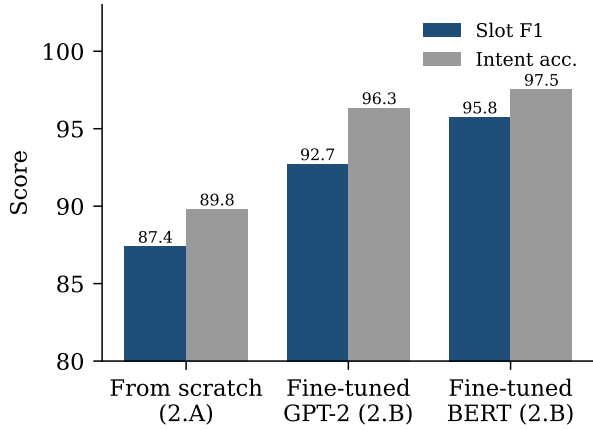


Figure 1: *The three models on ATIS. Pretraining lifts both metrics; bidirectionality (BERT over GPT-2) lifts slot F1 more than intent accuracy.*

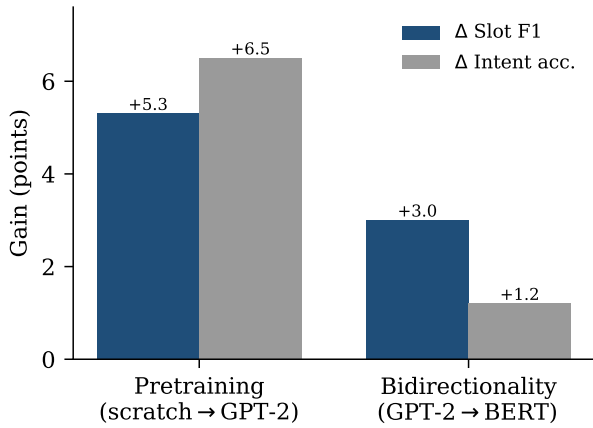


Figure 2: *Decomposing the gain over the from-scratch model. Pretraining is the larger mover; bidirectionality helps slots (+3.0) far more than intent (+1.2).*

Table 1: *ATIS test results, mean  $\pm$  standard deviation over five seeds. Slot F1 is computed with the conll scorer.*

| Model                  | Slot F1                            | Intent acc.                        |
|------------------------|------------------------------------|------------------------------------|
| From scratch (2.A)     | 87.40 $\pm$ 0.42                   | 89.81 $\pm$ 1.00                   |
| Fine-tuned GPT-2 (2.B) | 92.73 $\pm$ 0.26                   | 96.35 $\pm$ 0.45                   |
| Fine-tuned BERT (2.B)  | <b>95.75 <math>\pm</math> 0.21</b> | <b>97.54 <math>\pm</math> 0.28</b> |

Table 2: *Part 2.A incremental tuning. Each hyperparameter is changed on top of the previous configuration and kept only if the development metric improves.*

| Change                                           | Outcome  |
|--------------------------------------------------|----------|
| Learning rate (set to $3 \times 10^{-4}$ )       | baseline |
| Width ( $d_{\text{model}}$ 20 $\rightarrow$ 200) | kept     |
| Heads (1 $\rightarrow$ 4)                        | kept     |
| Layers (1 $\rightarrow$ 2)                       | reverted |
| Feed-forward width (20 $\rightarrow$ 512)        | kept     |
| Dropout layer                                    | reverted |